PHD PROGRAM

Applied Data Science & Artificial Intelligence

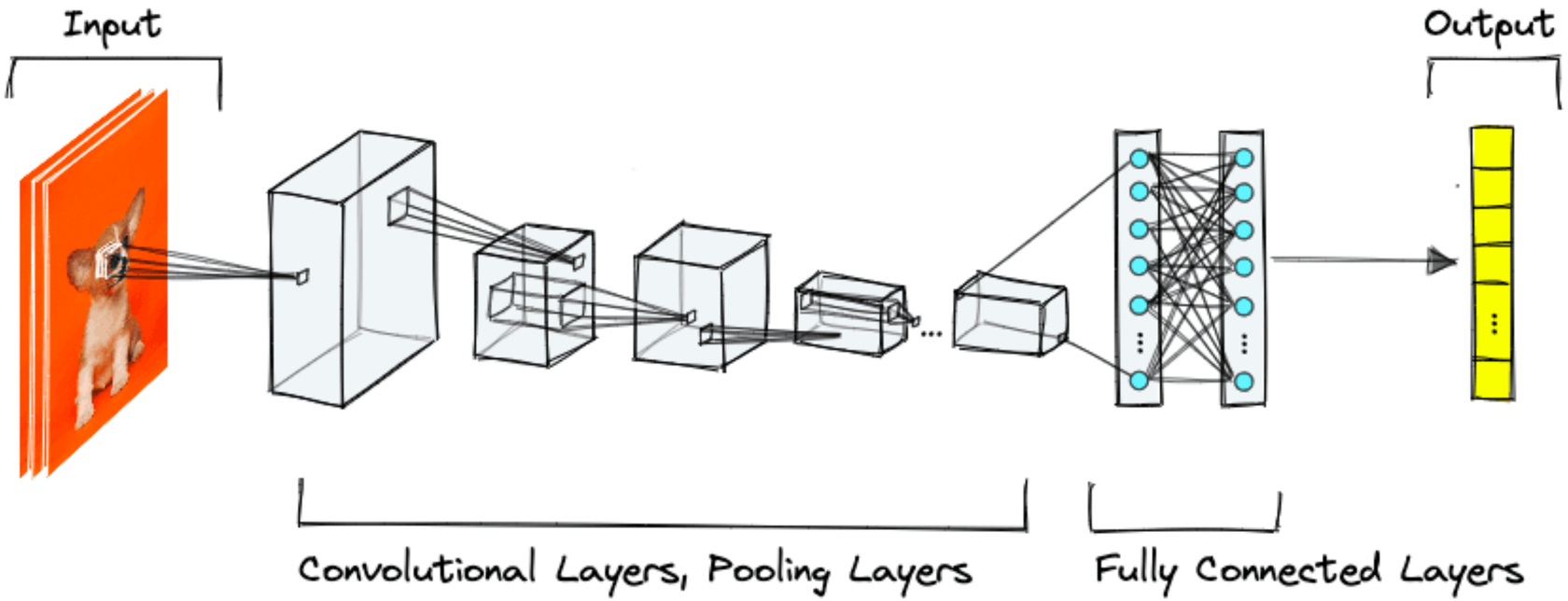
UNIVERSITÀ DEGLI STUDI DI TRIESTE

Explainable Artificial Intelligence

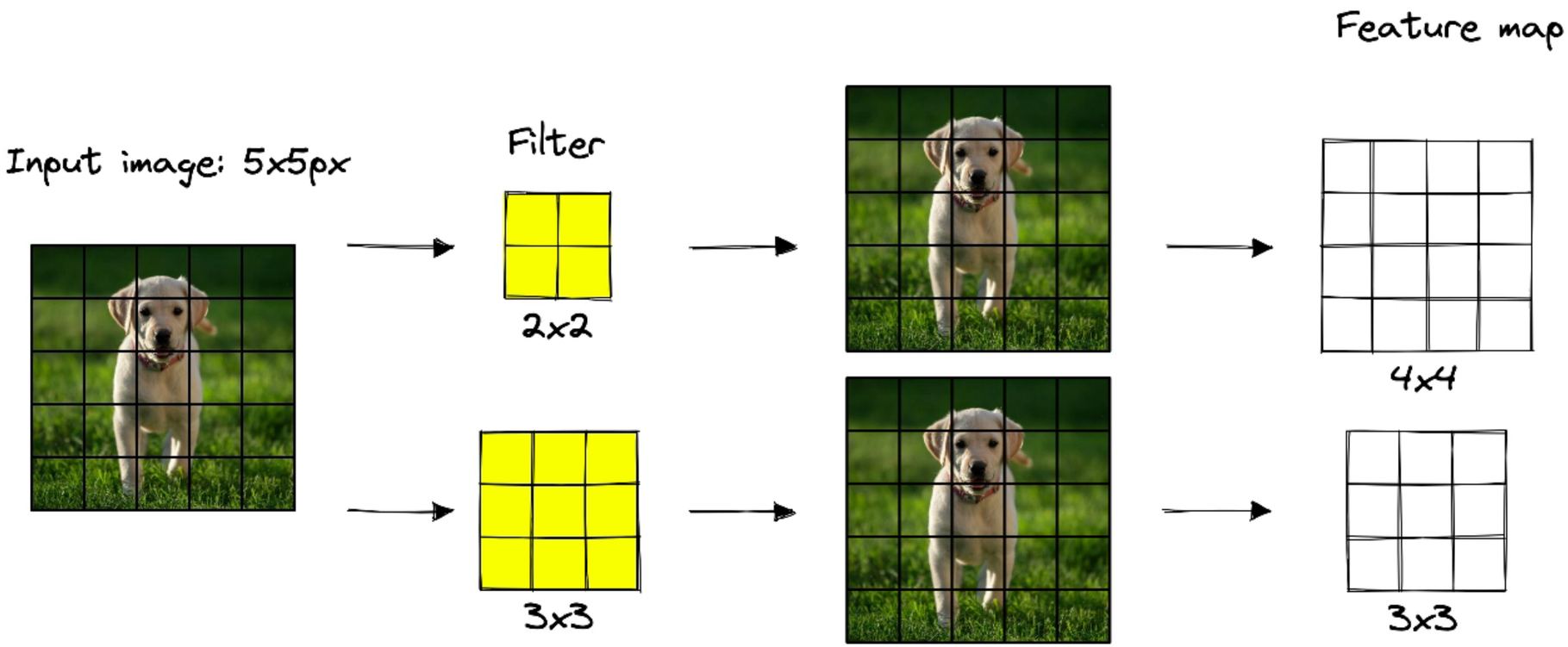
XAI for Convolutional Networks



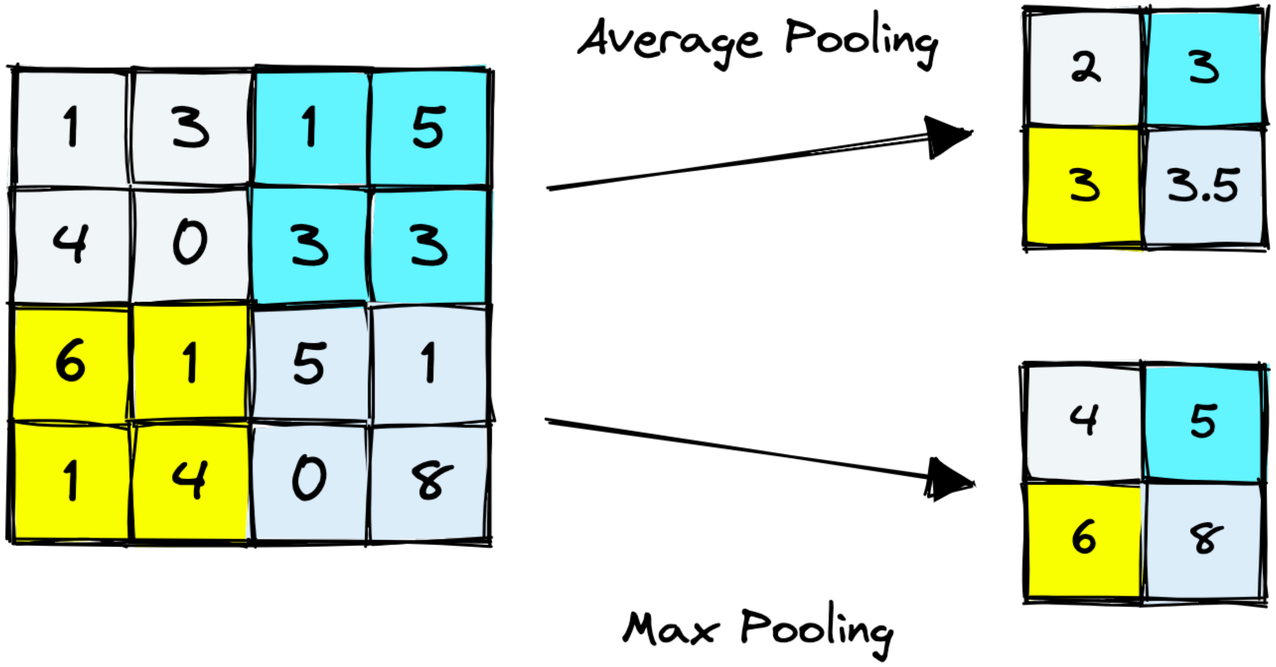
Introduction - Convolutional Neural Networks



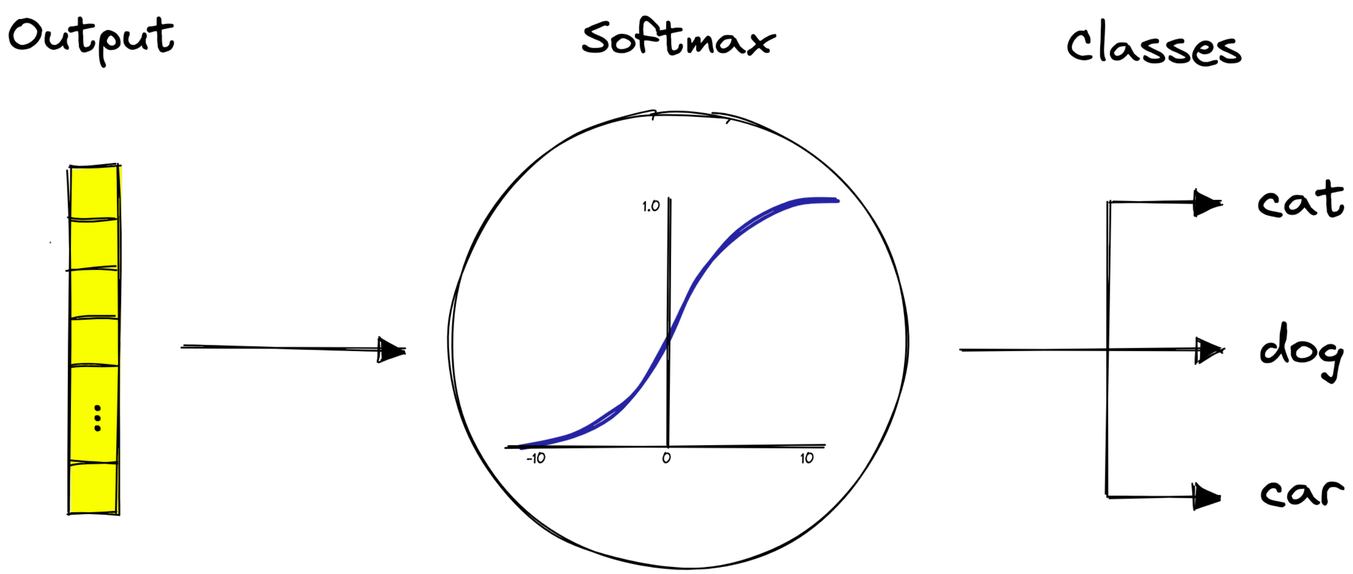
Introduction - Convolutional Layers



Introduction - Convolutional Neural Networks



Introduction - Convolutional Neural Networks



Introduction - The Challenge in Deep Learning Models

Challenges in Explaining CNNs:

- CNNs achieve great accuracy but are hard to interpret.
- **Feature maps in CNNs can show "what" is activated but not necessarily "why".**
- Need for techniques to visualize how networks make decisions.



(a) Normal



(b) Pneumonia

Visual Explainability

Advantages of Visualization Techniques (e.g., Grad-CAM):

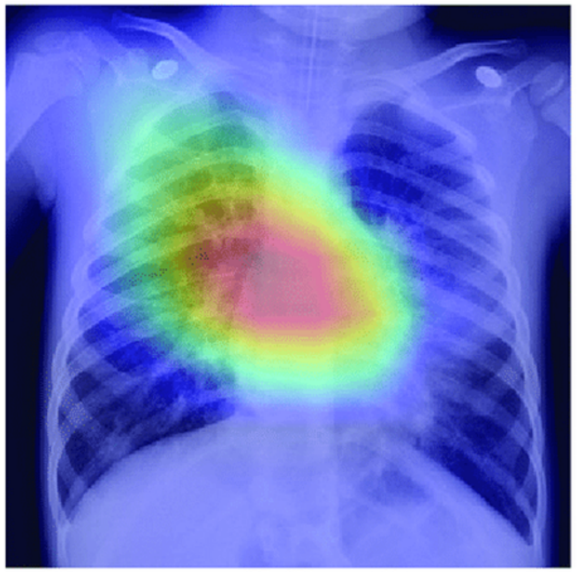
1. **Model Interpretability**:
-Offers intuitive and visual explanations, showing which regions in an image contribute to the prediction.
-Helps in understanding **why** a model made a specific decision, increasing trust.

2. **Improved Debugging**:
- Helps in identifying **bias** or **errors** in model predictions by revealing focus on irrelevant regions.
- Assists in model optimization by highlighting areas where the model may be focusing incorrectly.

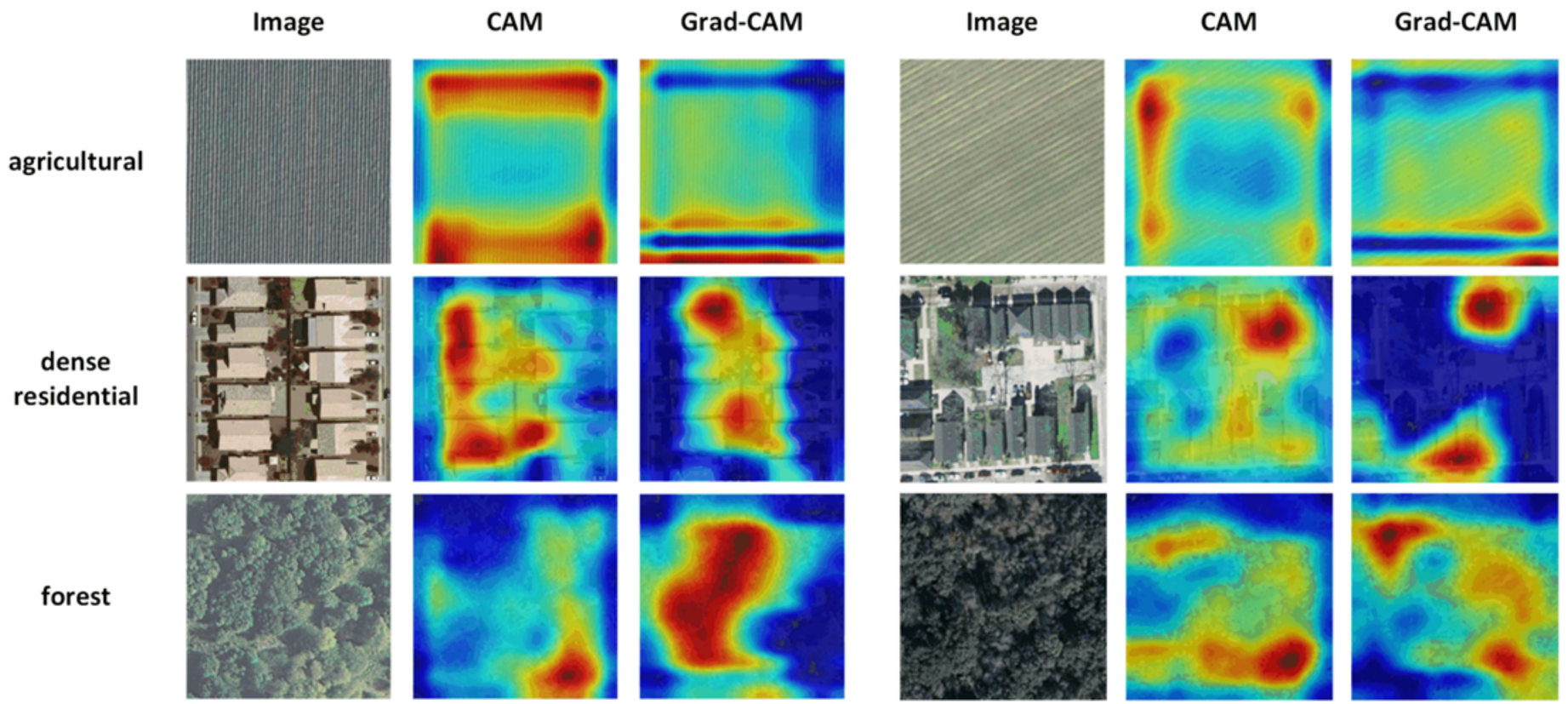
3. **Model Transparency**:
- Provides insights into what a deep learning model "sees," essential in sensitive domains like healthcare, finance, and law.

4. **Better Decision Making**:
- Helps experts (e.g., doctors) verify whether the model is making decisions based on relevant medical findings.

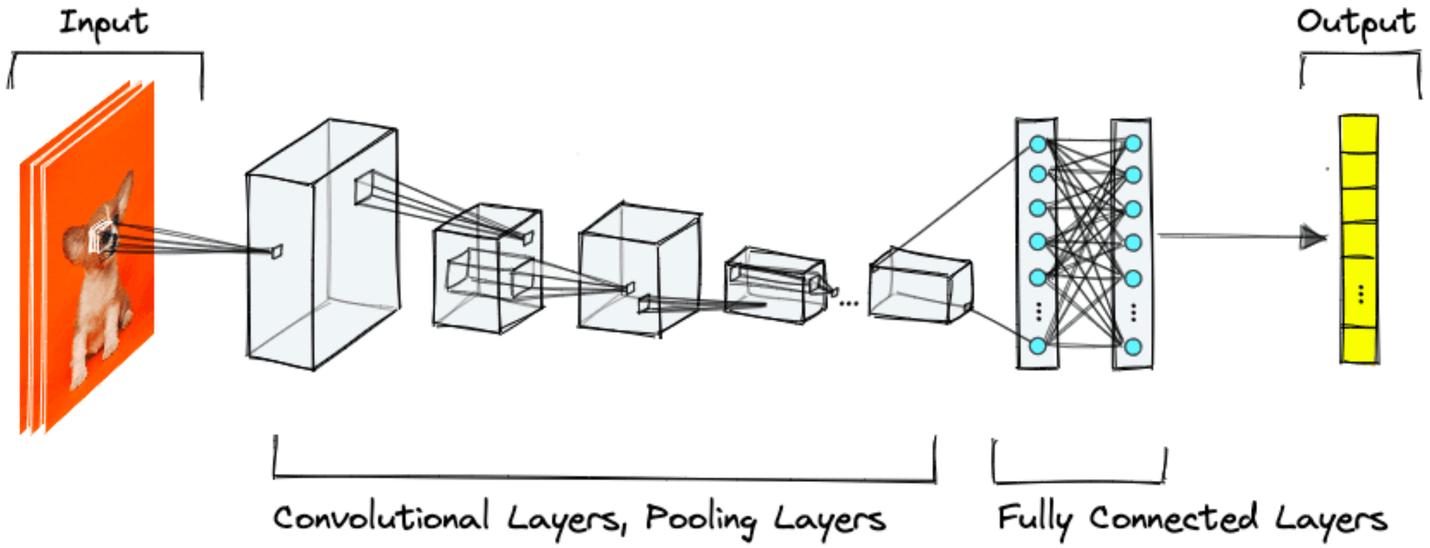




Exaplainable solutions



Intuition



Deeper representations within a CNN capture **higher-level visual constructs**. Additionally, convolutional layers inherently preserve **spatial information**, which is often lost in fully connected layers.
Hence the last convolutional layers provide an **optimal balance** between high-level semantics and detailed spatial information, as neurons in these layers are designed to **detect semantic class-specific information** within the image, such as object parts.

How do we aggregate the feature maps for different channels in the last convolutional layers?

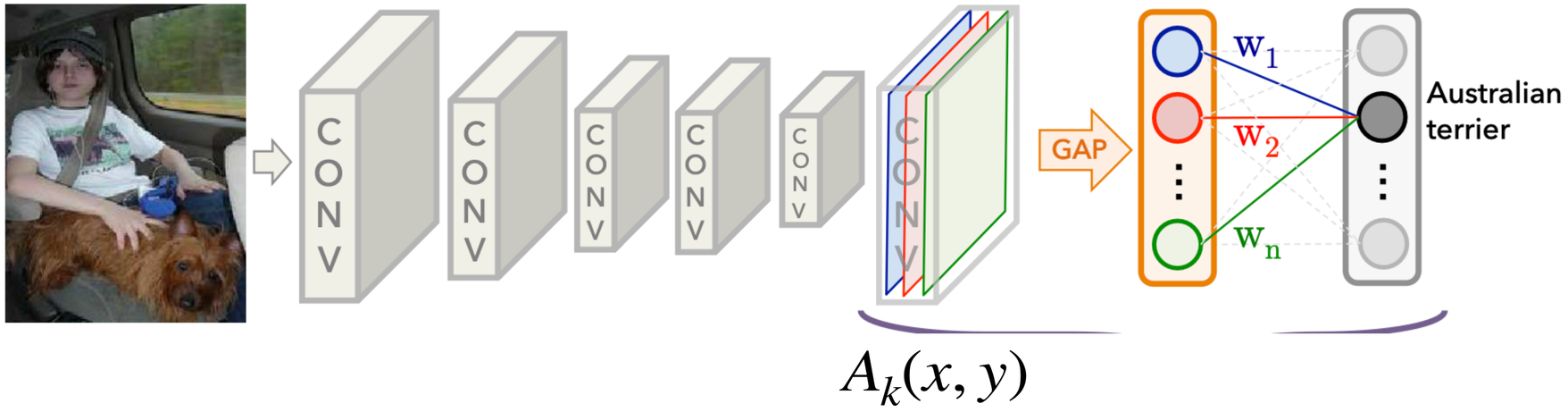
Class Activation Mapping (CAM)



CAM

Class Activation Mapping (CAM) is a method used to generate class-specific activation maps that highlight the critical regions of an input image, which are essential for classifying it into a particular category.

Zhou et al. (2016) introduced a CNN architecture that eliminates the need for fully connected layers by replacing them with Global Average Pooling, followed by a single softmax layer.



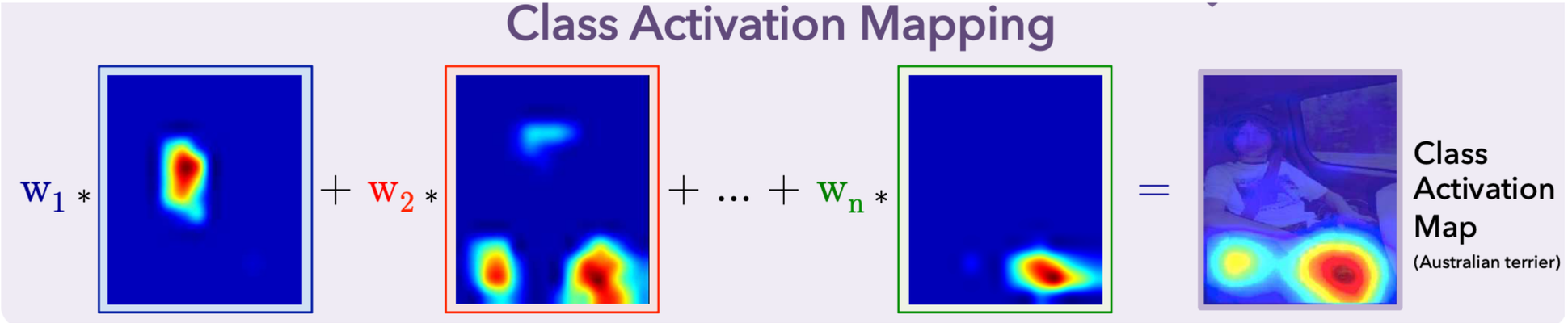
CAM

Rearranging terms, the pre-softmax activation for class c is:

$$y_c = \sum_k w_c^k \frac{1}{Z} \sum_{x,y} A_k(x,y) = \sum_{x,y} \frac{1}{Z} \sum_k w_c^k A_k(x,y) = \sum_{x,y} M_c(x,y)$$

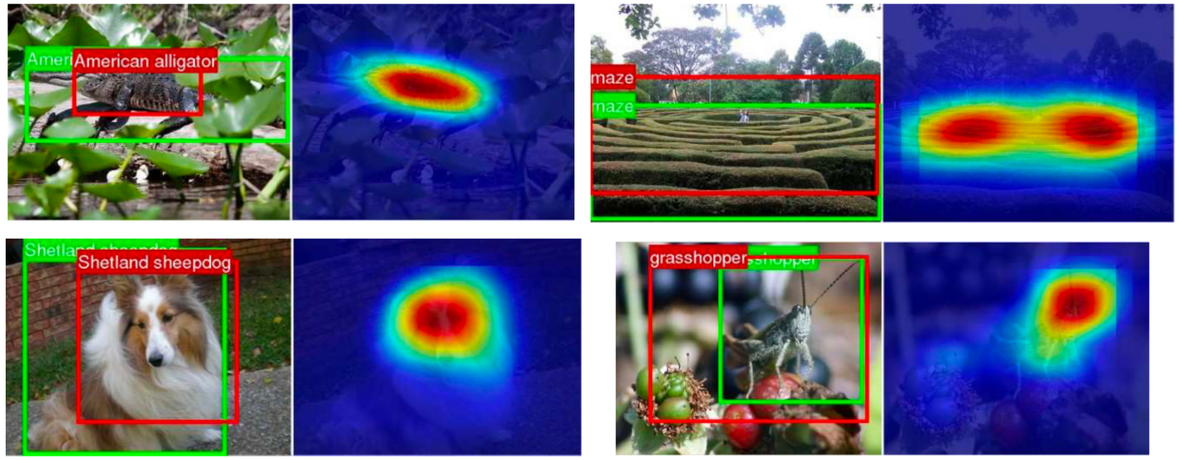
If we define the (normalized*) Class Activation Map Mc(x, y) as:

$$M_c(x,y) = \frac{1}{Z} \sum_k w_c^k A_k(x,y)$$



CAM

CAM is a very simple and effective method, and it achieved impressive results even when compared to object recognition benchmarks of the time.



However, there are some limitations:

- Class Activation Maps are ‘projected’ back into the pixel space by upsampling the feature maps from the final convolutional layer, resulting in a lack of fine, pixel-level detail.

- This technique was designed for a specific CNN architecture that uses Average Pooling, making it difficult to generalize to more traditional network architectures.

Grad-CAM



Grad-CAM

GradCAM (Selvaraju et al., 2016) is one of the best known follow-ups of CAM. It promises to solve the main issues of CAM, by providing a more general approach to class-specific activation Mapping.

Focus remains in the deepest convolutional layer, but the weighting scheme is changed to deal with generic architectures and non-linearities after the last convolutional layer.

Grad-CAM utilizes the gradient information flowing into the last convolutional layer of the CNN to assign importance values to each neuron for a specific decision of interest.

Grad-CAM

Grad-CAM works by following these key steps:

1. **Forward Pass**: The input image is passed through the convolutional neural network (CNN), and the model makes a prediction. During this pass, feature maps (activations) from a specific convolutional layer are stored.

2. **Backward Pass**: Gradients of the predicted class score (output) are calculated with respect to the feature maps of the selected convolutional layer. These gradients indicate how important each feature map is for the model's decision.

3. **Weight Calculation**: The gradients are averaged across all spatial locations to compute a weight for each feature map. These weights represent the importance of each feature map for the target class.

4. **Weighted Feature Maps**: The feature maps are then weighted by their corresponding importance scores, producing a weighted combination of the maps.

5. **Heatmap Generation**: The resulting weighted feature map is passed through a ReLU function (to focus on positive influences) and upsampled to match the original image size. This produces a heatmap that highlights the regions in the image that most influenced the model's decision.

Grad-CAM

This heatmap, overlaid on the original image, shows the areas of the input that contributed most to the model's classification, providing visual insights into the model's reasoning.



Grad-CAM

The activation map is defined as:

$$M_c(x, y) = ReLU(\sum_k \alpha_k^c A_k(x, y))$$

With:

$$\alpha_k^c = \frac{1}{Z} \sum_{x,y} \frac{\partial y_c}{\partial A_k(x, y)}$$

$A_k(x, y)$ is the k-th map of the last convolutional layer.

Grad-CAM

When used on the last layer in the context of the original CAM paper (all-conv network with global average pooling), GradCAM is exactly equivalent to CAM:

$$M_c^{GradCAM}(x,y) = \sum_k \alpha_k^c A_k(x,y) = \sum_k \frac{1}{Z}\left(\sum_{x,y} \frac{\partial y_c}{\partial A_k(x,y)}\right) A_k(x,y) = \sum_k \frac{1}{Z}\left(\sum_{x,y} \frac{w_c^k}{Z}\right) A_k(x,y) =$$

The pre-softmax yc has the form:

$$y_c = \sum_{x,y} \frac{1}{Z} \sum_k w_c^k A_k(x,y)$$

$$= \frac{1}{Z} \sum_k w_c^k A_k(x,y) = M_c^{CAM}(x,y)$$

And, consequently:

$$\frac{\partial y_c}{\partial A_k(x,y)} = \frac{w_c^k}{Z}$$

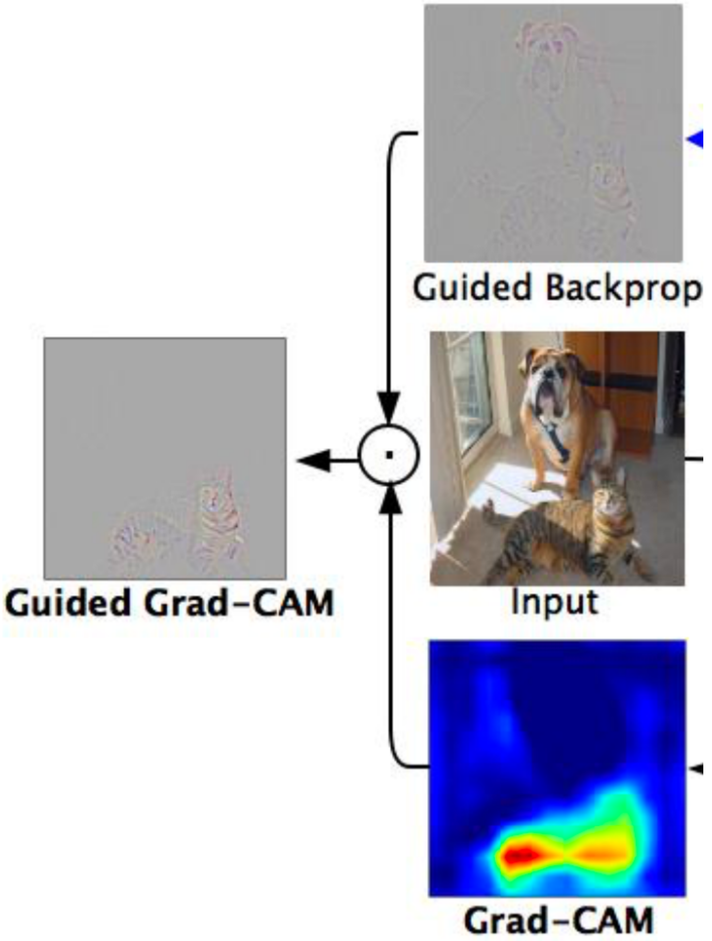
Guided Grad-CAM

While Grad-CAM is effective at distinguishing between classes and localizing important regions of an image, it falls short in capturing fine-grained details, unlike pixel-space gradient visualization methods that offer more precise, detailed insights.

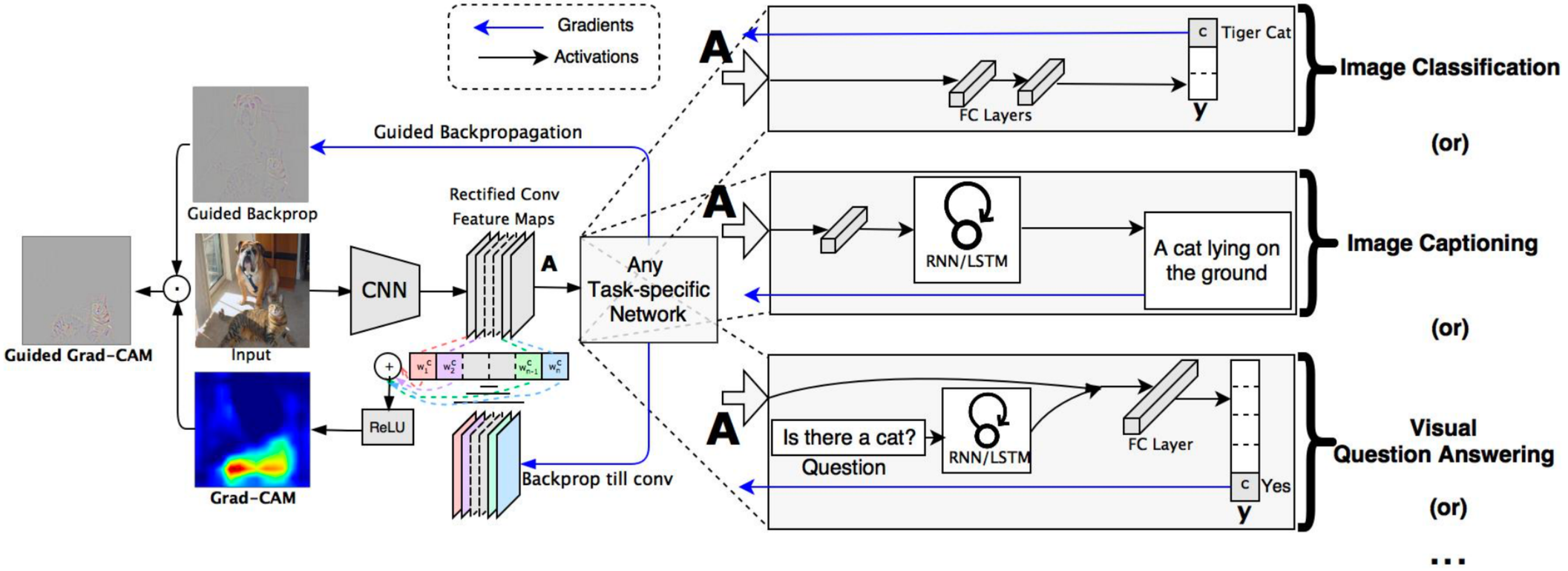
Guided Grad-CAM enhances the Grad-CAM heatmap by combining it with the detailed, pixel-level information from **Guided Backpropagation**, which visualizes the exact features (edges and textures) in the input image that contribute to the prediction.

- **Mechanism**:
  - First, Grad-CAM generates the coarse localization heatmap.
  - Then, Guided Backpropagation (which visualizes how individual pixels influence the output) is applied, resulting in sharper, more detailed visualizations.

- **Output**: The final result is a more precise and fine-grained visualization that highlights both *where* and *what* in the image influenced the prediction.



Guided Grad-CAM



In short, **Guided Grad-CAM** combines the **localization ability** of Grad-CAM with the **high-resolution detail** of Guided Backpropagation, giving more interpretable and visually rich explanations.

Grad-CAM examples



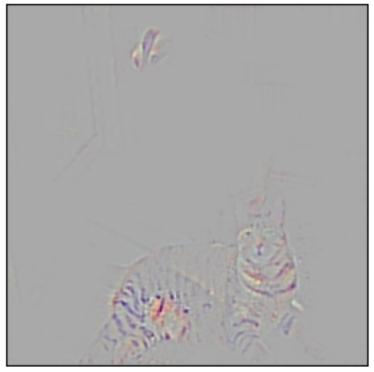
(a) Original Image



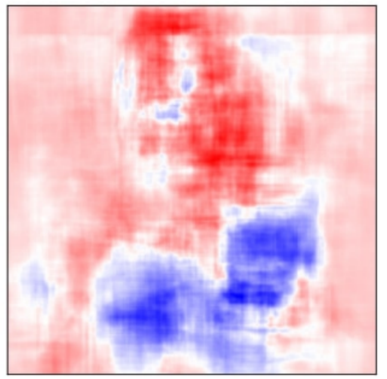
(b) Guided Backprop 'Cat'



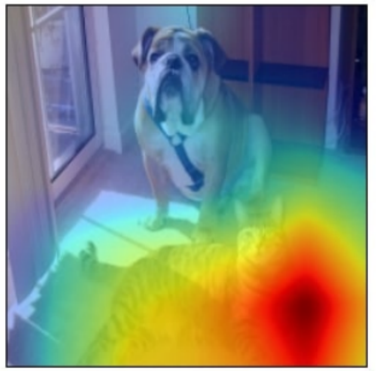
(c) Grad-CAM 'Cat'



(d)Guided Grad-CAM 'Cat'



(e) Occlusion map 'Cat'



(f) ResNet Grad-CAM 'Cat'

Grad-CAM examples: image captioning



Weight computed by the gradient of the log-probability of the caption w.r.t. the last convolutional layer $A_k(x, y)$

Grad-CAM examples: “counterfactual” explanations



(a) Original Image (b) Cat Counterfactual exp (c) Dog Counterfactual exp

Focus attention on evidence against class c.

$$\alpha_k^c = \overbrace{\frac{1}{Z}\sum_i\sum_j}^{\text{global average pooling}} \underbrace{-\frac{\partial y^c}{\partial A_{ij}^k}}_{\text{Negative gradients}}$$

Grad-CAM examples: adversarial attacks



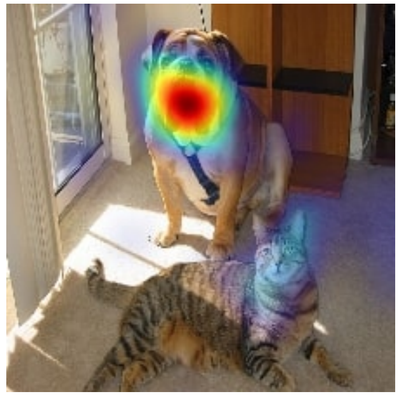
Boxer: 0.4 Cat: 0.2

(a) Original image



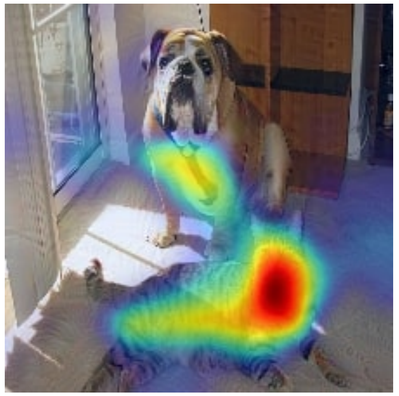
Airliner: 0.9999

(b) Adversarial image



Boxer: 1.1e-20

(c) Grad-CAM “Dog”



Tiger Cat: 6.5e-17

(d) Grad-CAM “Cat”



Airliner: 0.9999

(e) Grad-CAM “Airliner”



Space shuttle: 1e-5

(f) Grad-CAM “Space Shuttle”

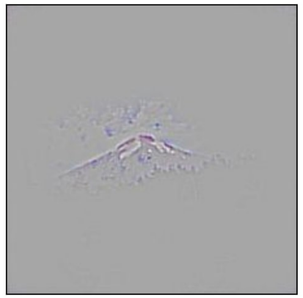
Grad-CAM examples: analysing failure modes



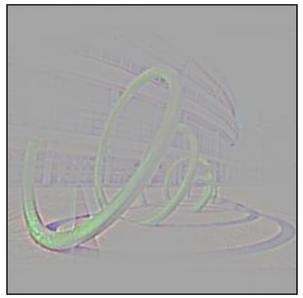


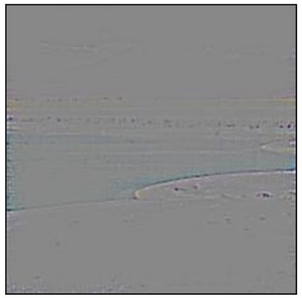


Ground truth: volcano

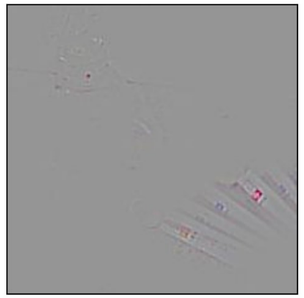

Ground truth: volcano

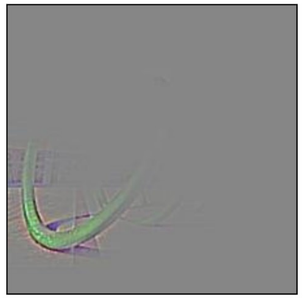

Ground truth: beaker


Ground truth: coil

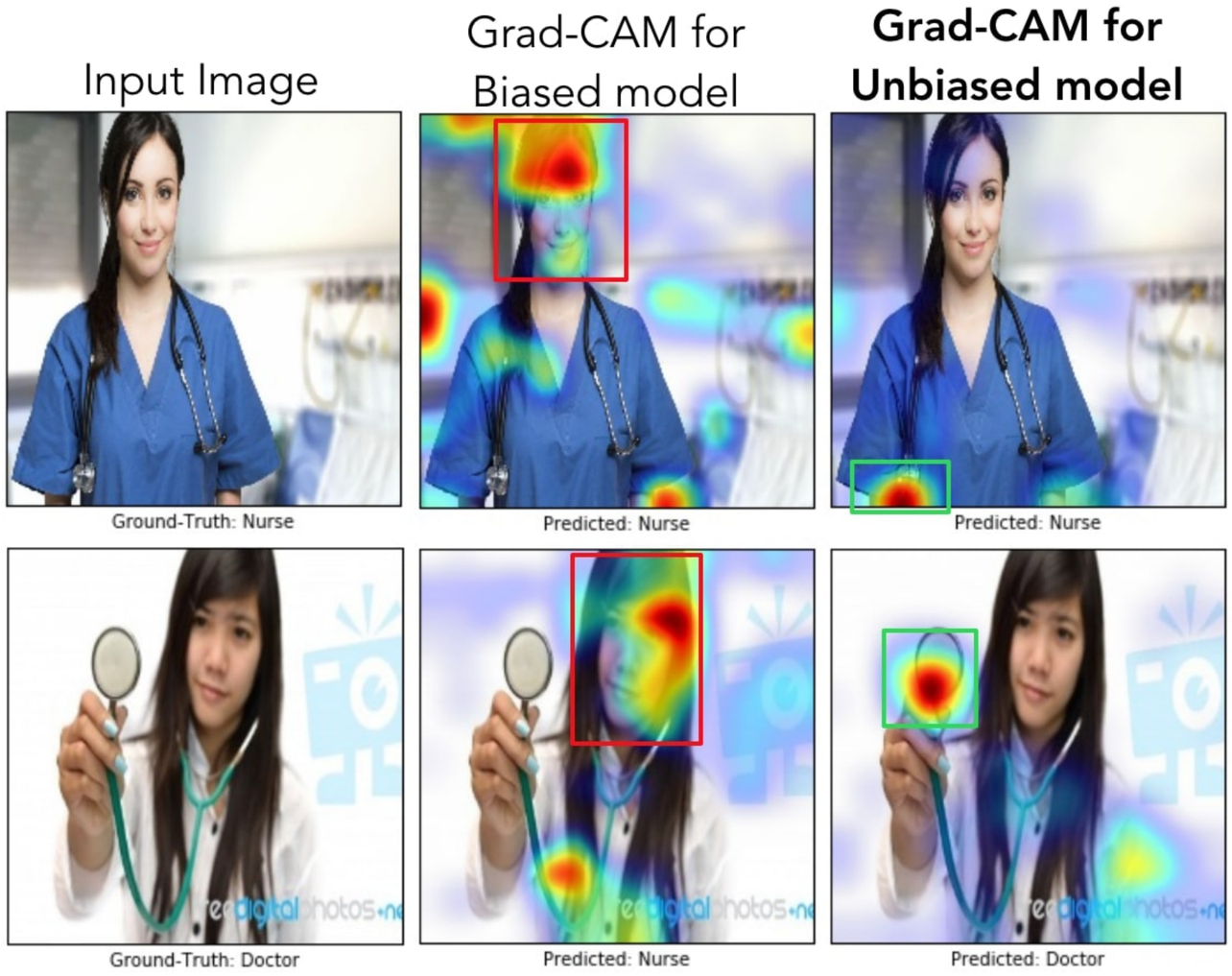

Predicted: sandbar


Predicted: car mirror

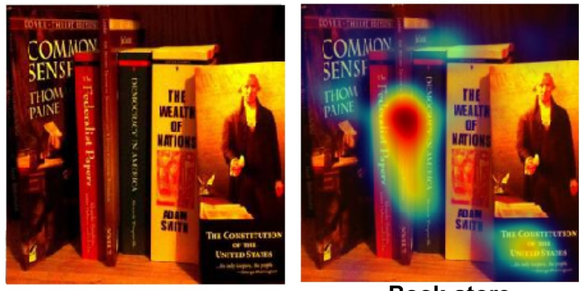

Predicted: syringe


Predicted: vine snake

Grad-CAM examples: checking for biases



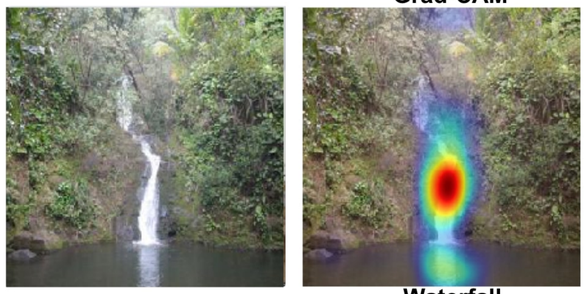
Grad-CAM examples



Important concepts for `**Book-store**`

Positive		Negative	
Neuron ID	Concept	Neuron ID	Concept
78	Book	237	Sky
318	Book	357	road
502	Striped	148	Water
311	Shelf	404	Car
156	Swirly	71	Flower

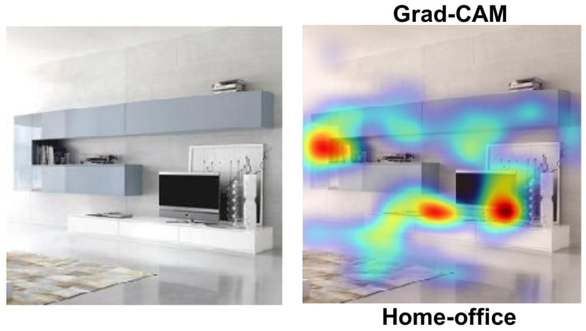
(a)



Important concepts for `**Waterfall**`

Positive		Negative	
Neuron ID	Concept	Neuron ID	Concept
117	Waterfall	115	Corridor
106	Closet	166	Road
148	Water	494	Bus
143	Water	106	Laundromat
216	Stratified	412	Grid

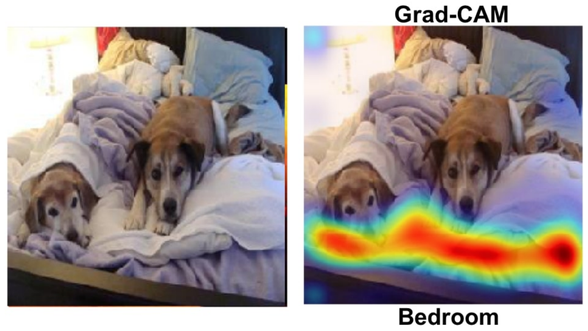
(b)



Important concepts for `**Home-office**`

Positive		Negative	
Neuron ID	Concept	Neuron ID	Concept
78	Book	186	Chequered
312	Desk	237	Sky
75	Office	494	Swimming pool
492	Stove	334	Sidewalk
305	Screen	498	Crosswalk

(c)



Important concepts for `**Bedroom**`

Positive		Negative	
Neuron ID	Concept	Neuron ID	Concept
317	Bed	187	Spiralled
290	Bed	294	Pantry
226	Painting	26	Toiled
175	Cushion	9	Shoe shop
117	Waterfall	182	Amusement park

(d)

Focus on the top-5 and bottom-5 neurons based on their class-specific importance scores, α_k. The names for these neurons can be used as text explanations.

D. Bau, B. Zhou, A. Khosla, A. Oliva, and A. Torralba. Network dissection: Quantifying interpretability of deep visual representations. In Computer Vision and Pattern Recognition, 2017.

Grad-CAM

Limitations of Grad-CAM:

1. **Coarse Localization**: Grad-CAM produces heatmaps that localize regions of interest but often at a coarse level. The heatmaps highlight general areas of the image rather than specific, fine-grained details, making it less effective when precise localization is needed (e.g., in medical diagnostics or when identifying small, subtle features).

2. **Handling Multiple Objects**: Grad-CAM struggles with images containing multiple objects or overlapping features. When multiple objects from different classes are present in the same image region, Grad-CAM tends to assign importance to broader areas, which can result in poor distinction between objects, limiting its ability to handle multi-label classification tasks effectively.

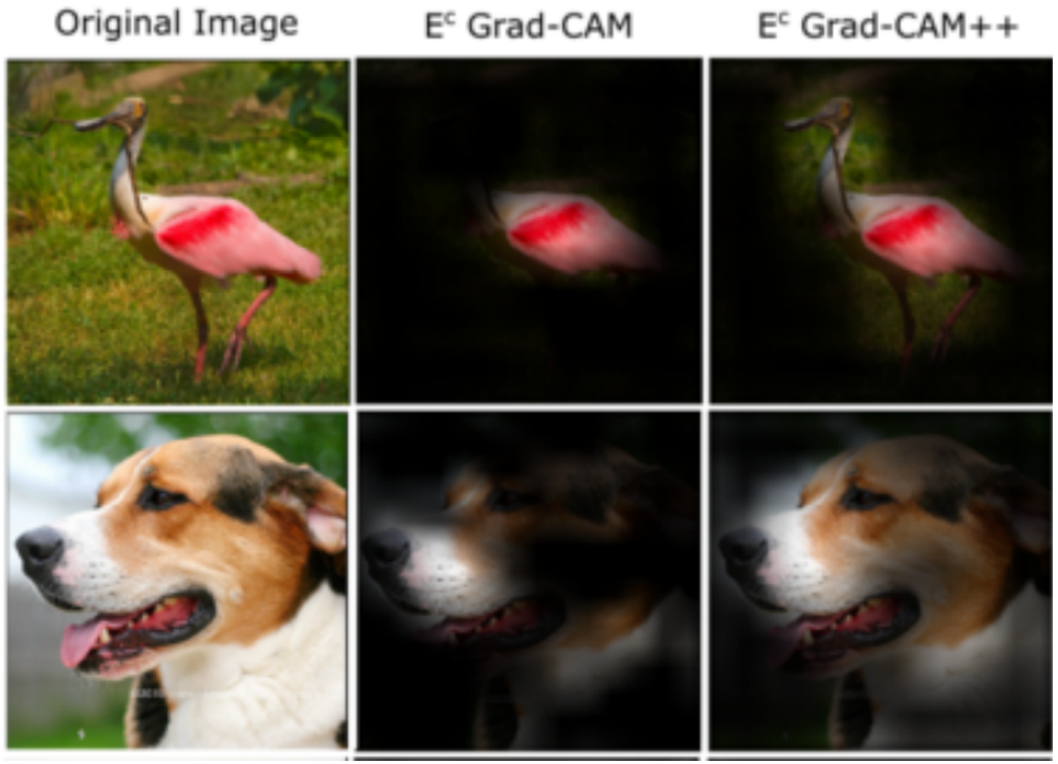
3. **Linear Assumptions**: Grad-CAM makes a simplifying assumption that the importance of a neuron's activation is linear, which can result in inaccurate importance scores when dealing with complex, non-linear interactions in convolutional layers. This can cause the model to misrepresent which parts of the image contribute to a decision.

Grad-CAM++



Grad-CAM++

Grad-CAM++ was developed to overcome Grad-CAM's limitations in **localization accuracy** and handling **multiple objects** or **overlapping class regions**, making it a more powerful tool for visualizing model decisions in complex scenarios.

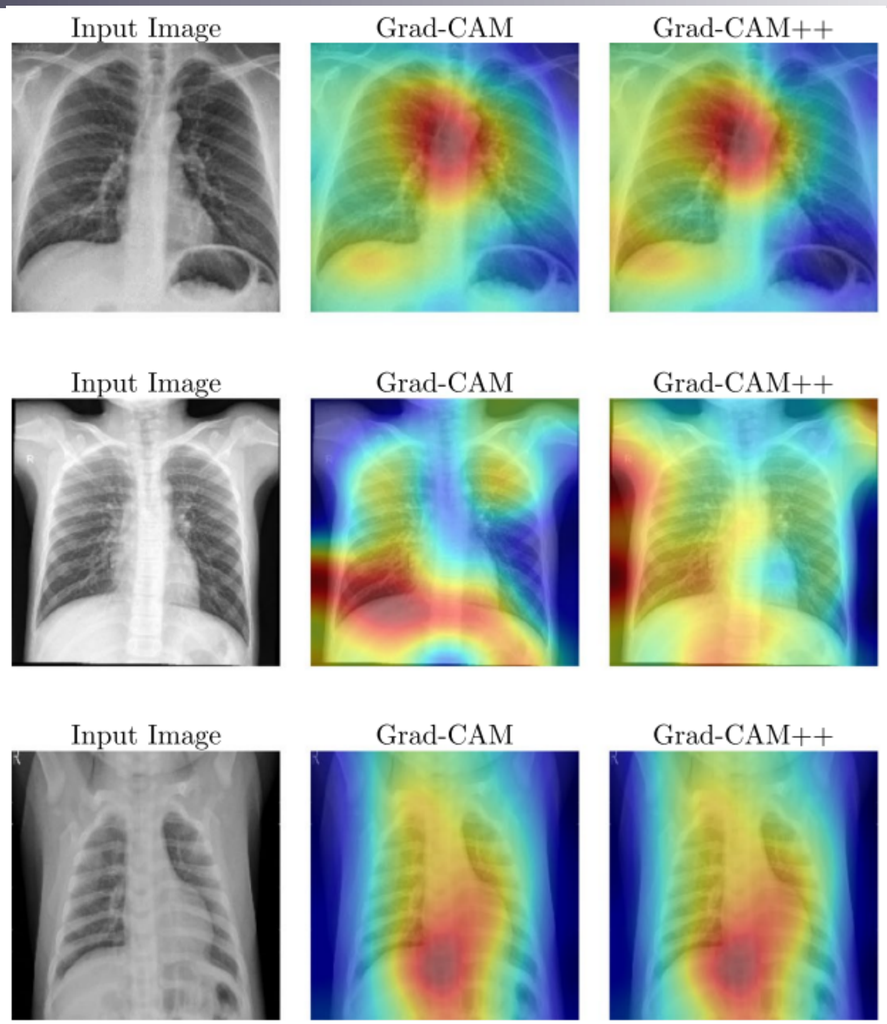


Grad-CAM++

- **Improved Localization**: Grad-CAM++ refines the way heatmaps are generated, leading to more precise, pixel-level localization of important regions. It addresses Grad-CAM's inability to capture subtle details by utilizing **higher-order derivatives**, which provide more accurate neuron importance scores.

- **Handling Multiple Objects and Classes**: Grad-CAM++ is designed to deal better with multi-object and multi-class scenarios by distributing importance more accurately across overlapping regions.

- **More Accurate Importance Scoring**: By using **higher-order gradients**, Grad-CAM++ accounts for non-linear dependencies between feature maps and output classes, leading to more accurate and reliable importance scoring.



Grad-CAM++: Step-by-step

1. Forward Pass Through the Network

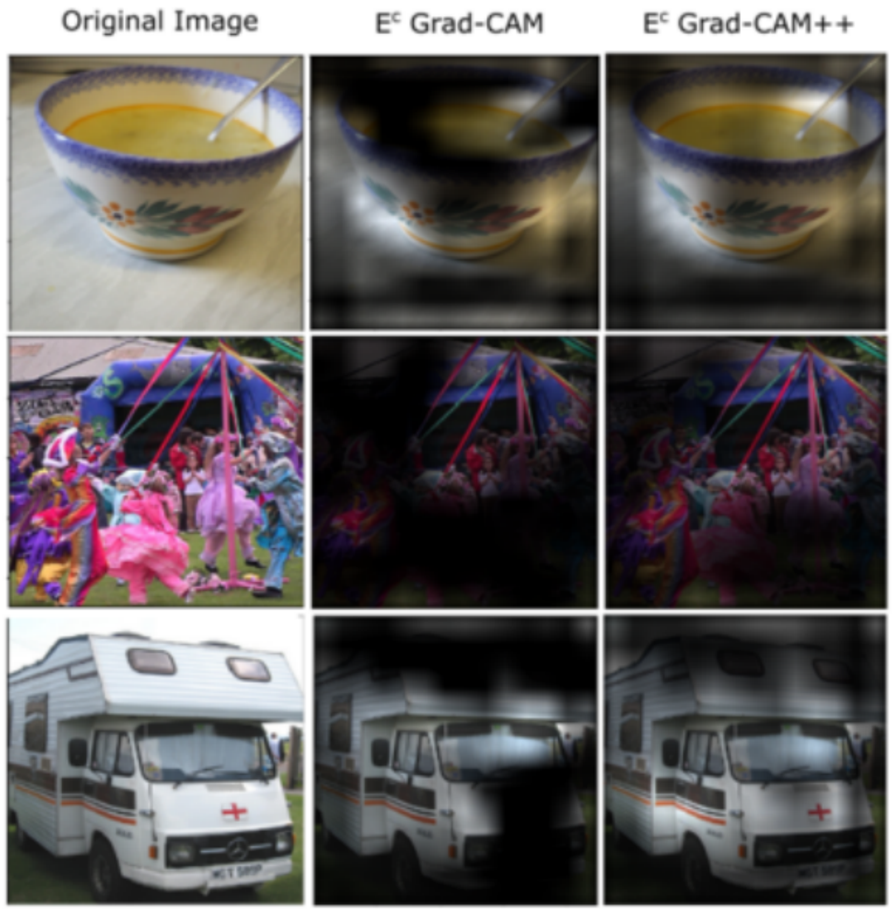
• Similar to Grad-CAM, the input image is passed through a Convolutional Neural Network (CNN), and feature maps (activations) are extracted from a specific convolutional layer.

2. Backward Pass to Compute Gradients

• Grad-CAM++ calculates the **first-order gradients** of the predicted class score with respect to the feature maps from the selected convolutional layer.
• **Grad-CAM++ also considers higher-order gradients**, capturing non-linear interactions between neurons in the feature maps and the target class.

3. Weighted Importance Computation

• It introduces **multiple weights per feature map**, computed using higher-order derivatives and are more sensitive to spatial locations and interactions between different parts of the image.
• The formula for Grad-CAM++ modifies the weighting process by introducing **positive contributions** of gradients at each pixel level> more accurate when dealing with multiple objects or subtle features.



Grad-CAM++: Step-by-step

4. Generating the Weighted Feature Maps

- The weights for each feature map are are multiplied by the corresponding feature maps from the forward pass to produce **weighted feature maps**.

- These feature maps are then summed up across all the channels to form a **class-discriminative map**.

5. Heatmap Generation with Pixel-Level Localization

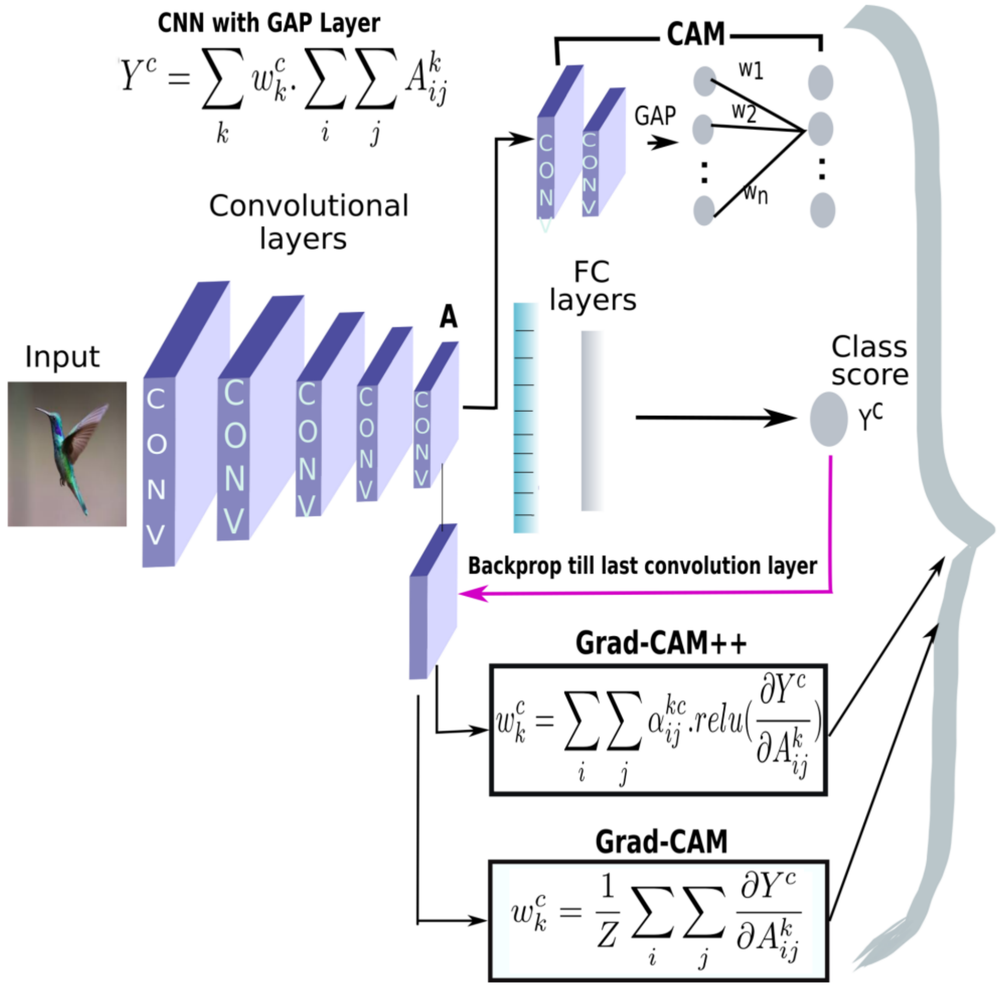
- The resulting weighted combination is passed through a **ReLU (Rectified Linear Unit)** function, ensuring that the heatmap only highlights regions with a positive contribution to the class prediction.

- Grad-CAM++ then upsamples this heatmap to the original image size, producing a **high-resolution, pixel-level heatmap**. This heatmap offers finer detail and sharper boundaries than Grad-CAM, allowing for more precise visualization.

6. Final Output: Refined Class Activation Map

- The heatmap generated by Grad-CAM++ is overlaid on the original image. This output offers better localization and sharper visualization of important image areas compared to Grad-CAM.

Grad-CAM++



Final Class discriminative Salinecy Map

$$L_{ij}^c = \sum_k w_k^c . A_{ij}^k$$

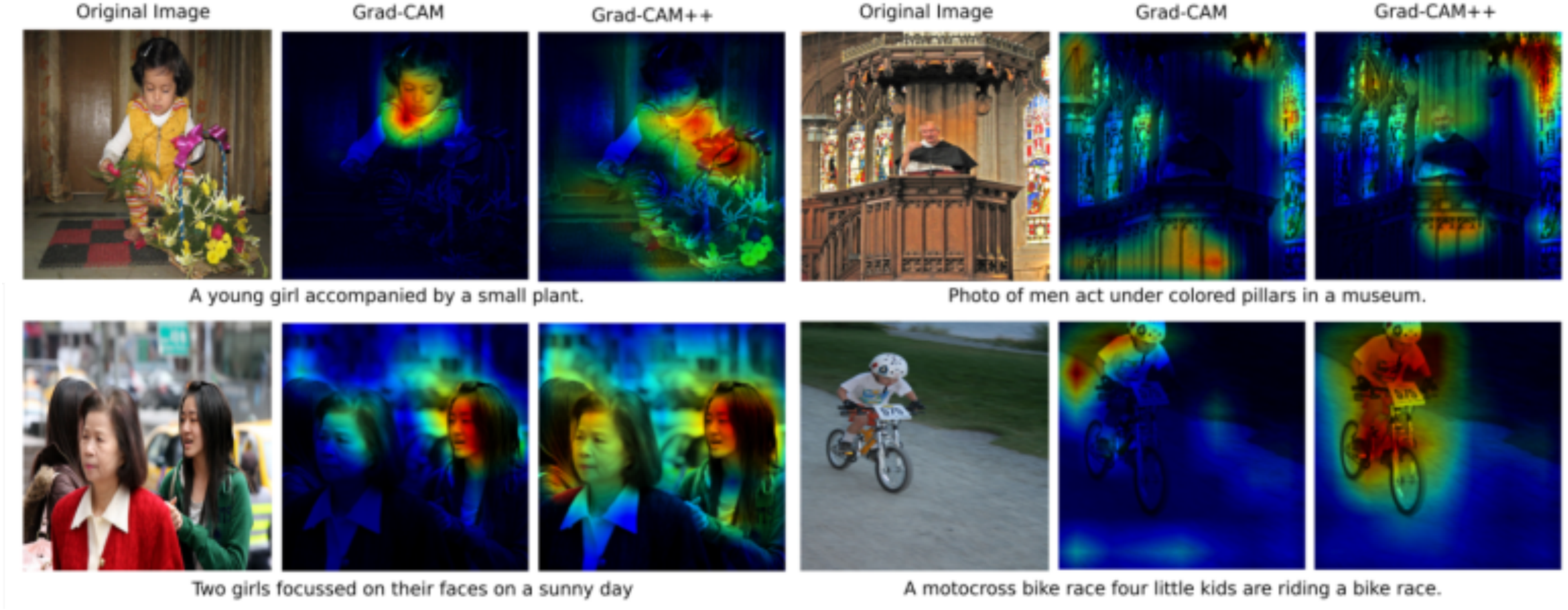
An overview of all the three methods – CAM, Grad-CAM, GradCAM++ – with their respective computation expressions.

GradCAM++ uses higher order derivatives to compute coefficients α_{ij}^{kc}.

$$\alpha_{ij}^{kc} = \frac{\frac{\partial^2 Y^c}{(\partial A_{ij}^k)^2}}{2\frac{\partial^2 Y^c}{(\partial A_{ij}^k)^2} + \sum_a \sum_b A_{ab}^k \{\frac{\partial^3 Y^c}{(\partial A_{ij}^k)^3}\}}$$

$$w_k^c = \sum_i \sum_j [\frac{\frac{\partial^2 Y^c}{(\partial A_{ij}^k)^2}}{2\frac{\partial^2 Y^c}{(\partial A_{ij}^k)^2} + \sum_a \sum_b A_{ab}^k \{\frac{\partial^3 Y^c}{(\partial A_{ij}^k)^3}\}}] . relu(\frac{\partial Y^c}{\partial A_{ij}^k})$$

Grad-CAM++: image captioning

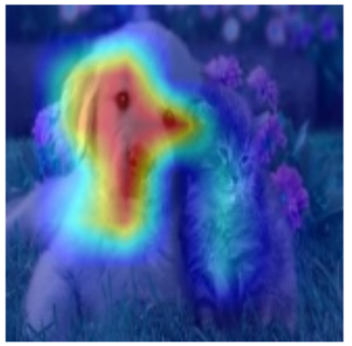


AblationCAM

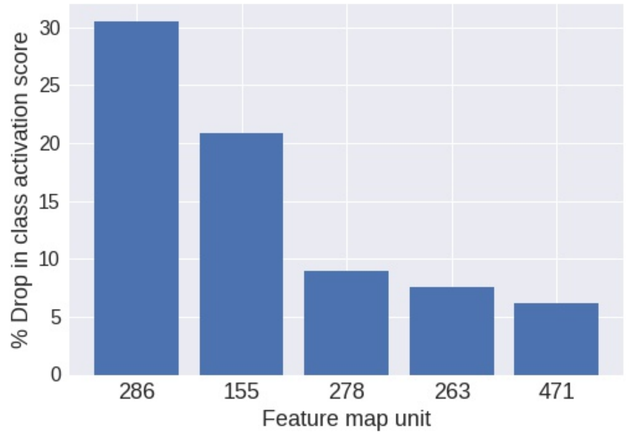


Ablation-CAM

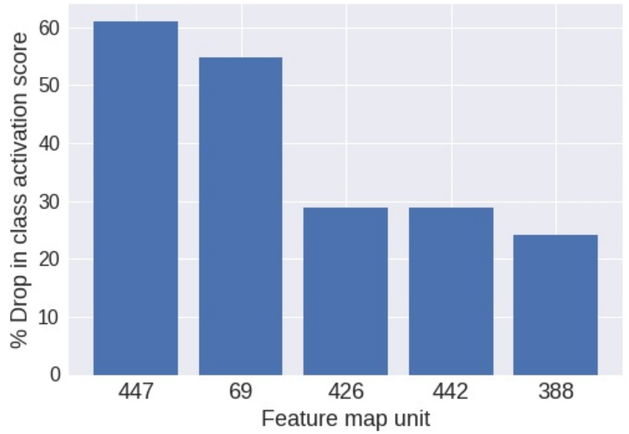
Ablation-CAM is a method for generating class activation maps (CAMs) that improves upon existing techniques like Grad-CAM.
It does this by systematically ablating (removing or disabling) parts of the network and observing how the output changes.

Category	Image	GradCAM	AblationCAM
Dog			
Cat			

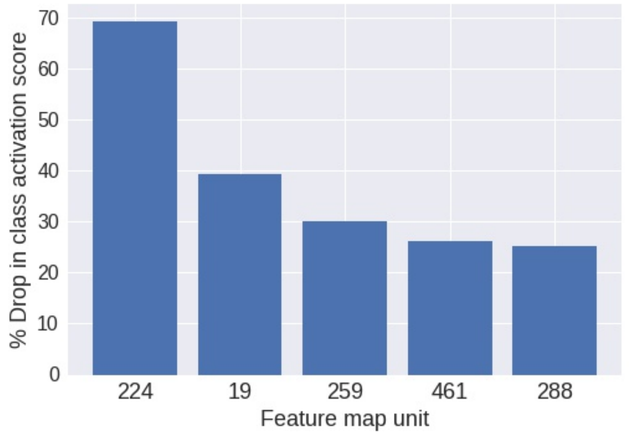
Ablation-CAM



(a) Leopard



(b) Grey Whale



(c) Digital watch

Figure 1: Activation score drop in decision nodes due to ablation of feature map units in final convolutional layer of VGG-16 network trained on Imagenet data for categories leopard(a), grey whale(b) and digital watch(c).

Removing different activation maps impacts differently on the activation scores of the target class

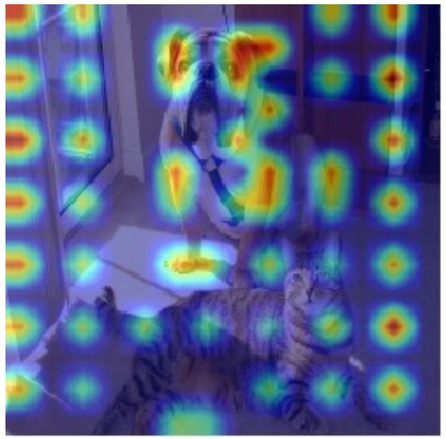
Ablation-CAM



(a) Original



(b) Unit 437 Visualization



(c) Gradient matrix visualization for unit 437

Pixel level gradients of the feature maps of deep convolutional layers are not class informative.

Figure 2: Visualizing feature map 437 (b) and corresponding gradient matrix (c) of VGG model's final convolutional layer for input image (a).

Ablation-CAM

Consider a case where we are required to generate a localization map for an image *I* using a CNN trained for image classification task.

A forward pass through the model is made to obtain the class activation score y^c of class c.

Lets assume this class score to be a non-linear function of feature map A_k of final convolutional layer, then y^c will be the value of this function when activations of A_k are present.

Set all the individual activation cell values of feature map A_k to zero and repeat the forward pass of same image *I*.

The ablation of unit k leads to a (possibly) reduced activation score y^c_k . Now, y^c_k is value of the function for absence of unit k and acts as a baseline for A_k.

Hence, the slope describing the effect of ablation of unit k is given by:

$$\text{slope} = \frac{y^c - y^c_k}{||A_k||}.$$

Ablation-CAM - Overview

The authors of Ablation-CAM argue that the slope used in their method offers a more reliable measure of feature importance compared to the "instantaneous slope" derived from gradients in Grad-CAM.

This approach is more robust, as it is **immune** to both **saturation**, where an **important filter may be incorrectly marked as unimportant**, and **explosion**, where a **filter with little value might be mistakenly assigned high importance**.

By avoiding these issues, **Ablation-CAM provides a more accurate representation of the significance of individual filters in the model's decision-making process.**

$$\text{slope} = \frac{y^c - y_k^c}{||A_k||}.$$

Ablation-CAM - Overview

The norm $\|A_k\|$ is often significantly larger than the numerator, causing the slope to take on a very small value.

This is overcome by modifying the slope to:

$$w_k^c = \frac{y^c - y_k^c}{y^c}$$

Ablation-CAM then computes as usual:

$$L^c_{Ablation-CAM} = \text{ReLU}\left(\sum_k w_k^c A_k\right)$$

Ablation-CAM - Examples



(a) Person class



(b) Grad-CAM



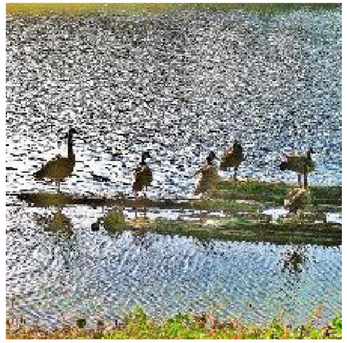
(c) Guided Grad-CAM



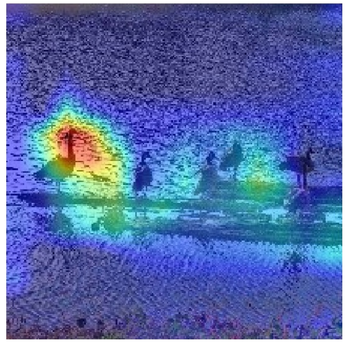
(d) Ablation CAM



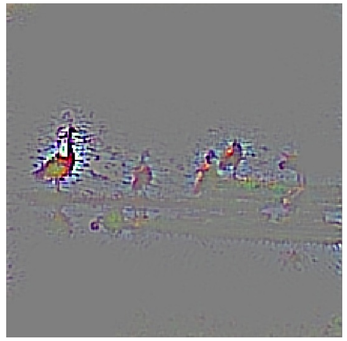
(e) Guided Ablation CAM



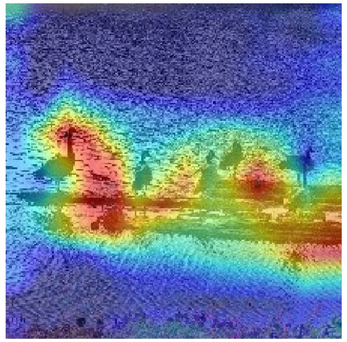
(f) Goose class



(g) Grad-CAM



(h) Guided Grad-CAM



(i) Ablation CAM



(j) Guided Ablation CAM

Ablation-CAM - Examples



(a) Bicycle : 1.0



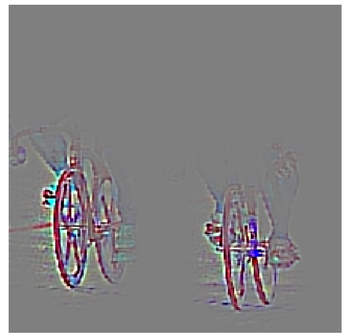
(b) Grad CAM



(c) Guided Grad CAM



(d) Ablation CAM



(e) Guided Ablation CAM



(f) Person : 1.0



(g) Grad CAM



(h) Guided Grad CAM



(i) Ablation CAM



(j) Guided Ablation CAM

References

Academic Papers:

- Zhou, Bolei, et al. "Learning deep features for discriminative localization." Proceedings of the IEEE conference on computer vision and pattern recognition. 2016. Link to Paper
- Selvaraju, Ramprasaath R., et al. "Grad-CAM: visual explanations from deep networks via gradient-based localization." International journal of computer vision 128 (2020): 336-359. Link to Paper
- Grad-CAM++: Improved Visual Explanations for Deep Convolutional Networks Aditya Chattopadhyay, Anirban Sarkar, Prantik Howlader, Vineeth N Balasubramanian. Link to Paper
- Saurabh Desai and Harish G Ramaswamy. "Ablation-cam: Visual explanations for deep convolutional network via gradient-free localization. ". In WACV, pages 972–980, 2020. Link to Paper

Online Resources:

- Implementation of Grad-cam for pytorch

ANY QUESTIONS?

THIS IS THE END



Thanks for the attention!